Statistical Learning and Data Analysis

Assignment 1: Exploratory Data Analysis (EDA)

You can work in a team (up to 3 members, with an heterogeneous composition in terms of bachelor degree and country of provenience). For the submission of the Assignment 1 please follow the instructions at the link: **https://forms.gle/8qfUfk9FcDfUcDdw8**

DATASET AND GOAL

You are provided with a selection of datasets, pick one.
Your goal is to perform a **complete Exploratory Data Analysis (EDA)** by answering the following questions step by step.
Each section builds upon the previous one.

OUTPUT REQUIREMENTS

The assignment must be submitted with a file containing code, plots and brief interpretations: all plots should have clear titles, labels and legends (when required) and each section should include at least one short commentary interpreting the results. The same file should be presented in the following formats:

- **.ipynb** or **.Rmd**
- **.pdf** or **.html**

Moreover, a short presentation should also be produced for the exam dissertation. The presentation should not be a copy of the code notebook. It should focus on interpreting the results and communicating them clearly, with clean visuals and minimal text instead.

1) DATA LOADING & CLEANING

- Import the dataset and display the first few rows. What are the variable names and their data types (numeric, categorical, etc.)?
- How many observations and variables are there?
- Are there any missing values? In which columns and in what proportion?
- How would you handle the missing values (removal, imputation, …)?

2) UNIVARIATE DATA DESCRIPTION

- Compute descriptive statistics (mean, median, min, max, standard deviation, etc…) for all numeric variables.
- Create frequency tables for categorical variables.
- Plot histograms and boxplots for numeric variables. What can you say about the distribution shape?
- Plot bar charts or pie charts for categorical variables. Which category is the most frequent?

3) BIVARIATE AND MULTIVARIATE DESCRIPTION

- Compute correlations among numeric variables and visualize them with a correlation heatmap.
- Create scatterplots for pairs of numeric variables. Do you observe linear or nonlinear relationships?
- Compare the distributions of a numeric variable across categories.
- If there are multiple categorical variables, build a contingency table.

4) PRINCIPAL COMPONENT ANALYSIS AND CLUSTERING

- Scale and encode the variable, then apply a Principal Component Analysis.
- How many components are required to explain at least 80% of the variance?
- Visualize the loading vectors on a correlation circle, and the observations on a biplot. What patterns or grouping do you observe? What the two principal components represent?
- Select a clustering algorithm
- Determine the optimal number of clusters (using the desired method)
- Visualize the resulting clusters. Are they well separated.

5) SUMMARY AND INTERPRETATION

- Summarize in a few sentences the main characteristics of the dataset.
- Have you identified any interesting patterns, or differences through the analysis.